

HW 4:

Exercises Section 7.2.2: Problems 5 and 7.HW4_questions.pdf

HW4

Section 7.2.2

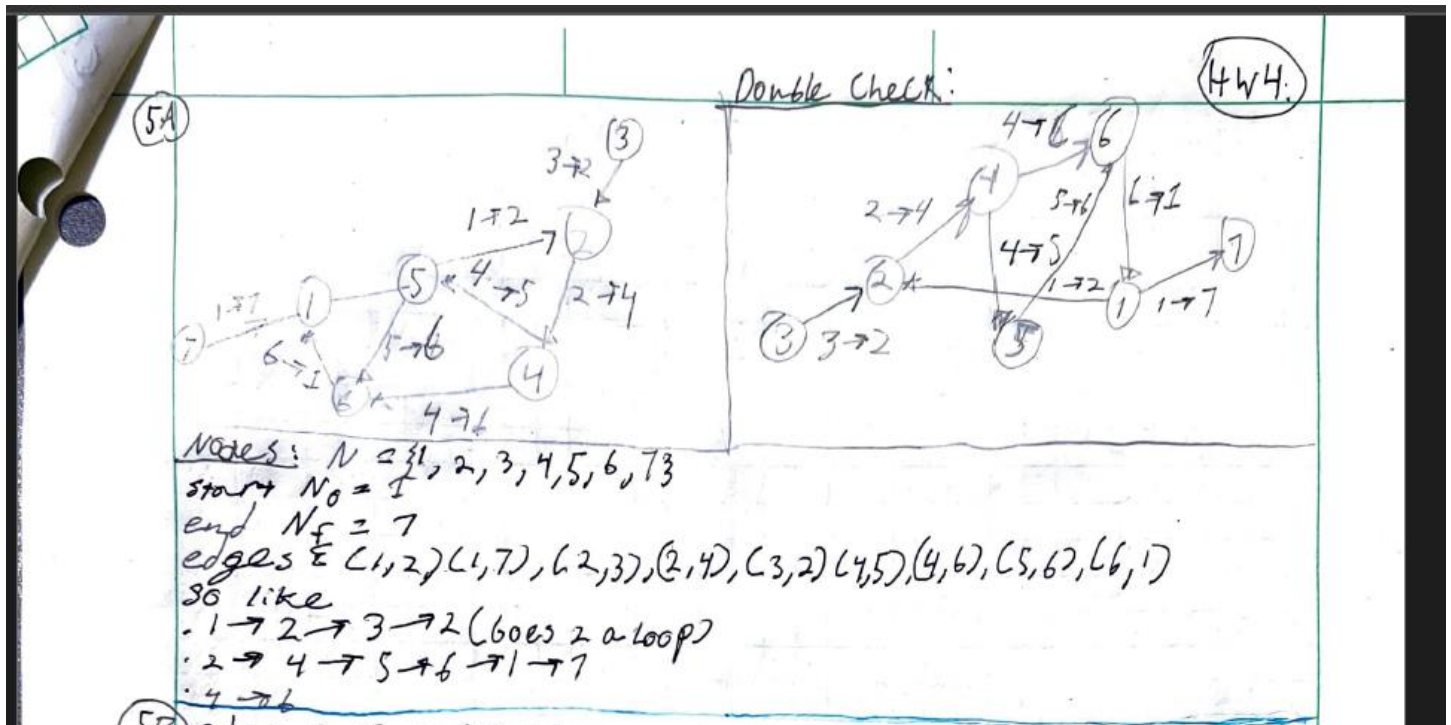
5. Answer questions a–g for the graph defined by the following sets:

- $N = \{1, 2, 3, 4, 5, 6, 7\}$
- $N_0 = \{1\}$
- $N_f = \{7\}$
- $E = \{(1, 2), (1, 7), (2, 3), (2, 4), (3, 2), (4, 5), (4, 6), (5, 6), (6, 1)\}$

Also consider the following (candidate) test paths:

- $p_1 = [1, 2, 4, 5, 6, 1, 7]$
- $p_2 = [1, 2, 3, 2, 4, 6, 1, 7]$
- $p_3 = [1, 2, 3, 2, 4, 5, 6, 1, 7]$

(a) Draw the graph.



- (b) List the test requirements for Edge-Pair Coverage. (Hint: You should get 12 requirements of length 2.)

4 → 6

5B edge Pair Coverage

1. (1, 2) → (2, 3)

2. (1, 2) → (2, 4)

3. (1, 7) ← No pairing

4. (2, 3) → (3, 2)

5. (2, 4) → (4, 5)

6. (2, 4) → (4, 6)

7. (3, 2) → (2, 3)

8. (3, 2) → (2, 4)

9. (4, 5) → (5, 6)

10. (4, 6) → (6, 1)

11. (5, 6) → (6, 1)

12. (6, 1) → (1, 2)

13. (6, 1) → (1, 7)

It's 13 Not 12

Paths of length 2
Double Check:

1. (1, 2) - (2, 3)

2. (1, 2) - (2, 4)

3. (2, 3) - (3, 2)

4. (2, 4) - (4, 5)

5. (2, 4) - (4, 6)

6. (3, 2) - (2, 3)

7. (3, 2) - (2, 4)

8. (4, 5) - (5, 6)

9. (4, 6) - (6, 1)

10. (5, 6) - (6, 1)

11. (6, 1) - (1, 2)

12. (6, 1) - (1, 7)

That's better.

5C So From Previous Question (5B) / 12

- (c) Does the given set of test paths satisfy Edge-Pair Coverage? If not, state what is missing.

So From Previous Question (5B) (Let's do reasoning)
 $P_1 = [1, 2, 4, 5, 6, 1, 7]$
 $P_2 = [1, 2, 3, 2, 4, 6, 1, 7]$
 $P_3 = [1, 2, 3, 2, 4, 5, 6, 1, 7]$

see next pg

	P_2	P_3	HW 4P
(5B) So P_1			
✓ (1,2) - (2,3)	✓ (1,2) - (2,3)	(1,2) - (2,3) ✓	
✓ (1,2) - (2,4)	✓ (2,3) - (3,2)	(2,3) - (3,2) ✓	
✓ (2,4) - (4,5)	✓ (3,2) - (2,4)	(3,2) - (2,4) ✓	
✓ (4,5) - (5,6)	✓ (2,4) - (4,5)	(2,4) - (4,5) ✓	
✓ (5,6) - (6,1)	✓ (4,5) - (5,6)	(4,5) - (5,6) ✓	
✓ (6,1) - (1,7)	✓ (6,1) - (1,7)	(5,6) - (6,1) ✓	
		(6,1) - (1,7) ✓	
We're missing (3,2) - (2,3) & (6,1) - (1,2) - Definitely NOT satisfied. Given paths do not satisfy edge-pair coverage.			
(5D)	Simple Path: [3, 2, 4, 5, 6] Test Path [1, 2, 3, 2, 4, 6, 1, 2, 4, 5, 6, 1, 7]		

- (d) Consider the simple path [3, 2, 4, 5, 6] and test path [1, 2, 3, 2, 4, 6, 1, 2, 4, 5, 6, 1, 7]. Does the test path tour the simple path directly? With a sidetrip? If so, write down the sidetrip.

5.1) satisfy edge-pair coverage. DO NOT

Simple Path: [3, 2, 4, 5, 6] Test Path [1, 2, 3, 2, 4, 6, 1, 3, 4, 5, 6, 1]

we have to check if the simple paths in sequence in test path so. $3 \rightarrow 2 \rightarrow 4 \rightarrow 5 \rightarrow 6$

so looking at graph & previous full test path = $\rightarrow$ in between

Answers:

Index: 2: 3, 3: 2, 4: 4, 4: 6 which is not (5)

7: 2, 8: 4, 9: 5, 10: 6

so there's side trip of $4 \rightarrow 6 \rightarrow 1 \rightarrow 2 \rightarrow 4$ before return

The test path does tour w/ a path [3, 2, 4, 5, 6]

side trip path: [4, 6, 1, 2, 4]

5.2) Test (NCC, EL, PPL) AKA Node, edge, & Prime Path

(e) List the test requirements for Node Coverage, Edge Coverage, and Prime Path Coverage on the graph.

side trip path: [4, 6, 1, 2, 4]

5.3) So, Test (NCC, EL, PPL) AKA Node, edge, & Prime Path Coverage

① NCC from $N = \{1, 2, 3, 4, 5, 6, 7\}$

TR = $\{1, 2, 3, 4, 5, 6, 7\}$

② EL: $E = \{(1, 2), (1, 7), (2, 3), (2, 4), (3, 2), (4, 5), (4, 6), (5, 6), (6, 1)\}$

Traverse every edge at least 4's.

③ Prime paths are not simple paths

Prime = Non a sub path of a longer simple path

simple = no repeated nodes

length 1: $\{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}, \{7\}$

length 2: $\{1, 2\}, \{1, 7\}, \{2, 3\}, \{2, 4\}, \{3, 2\}, \{4, 5\}, \{4, 6\}, \{5, 6\}, \{6, 1\}$

longer prime paths:

[1, 2, 3, 2]	[2, 4, 6, 1]
[1, 2, 4, 5]	[4, 5, 6, 1]
[1, 2, 4, 6]	[5, 6, 1, 2]
[2, 3, 2, 4]	[6, 1, 2, 3]
[3, 2, 4, 5]	[6, 1, 7]
[3, 2, 4, 6]	
[2, 4, 5, 6]	

prime path coverage

TR

(f) List test paths from the given set that achieve Node Coverage but not Edge Coverage on the graph.

4. f) ① Node / edge sets

$N = \{1, 2, 3, 4, 5, 6, 7\}$

$E = \{(1, 2), (1, 7), (2, 3), (2, 4), (3, 2), (4, 5), (4, 6), (5, 6), (6, 1)\}$

② Check test paths

$P_1 = [1, 2, 4, 5, 6, 1, 7]$

Nodes hit: 1, 2, 4, 5, 6, 7 $\leftarrow$ don't have 3

used edges: $\{(1, 2), (2, 4), (4, 5), (5, 6), (6, 1), (1, 7)\}$

Missing edges: $\{(2, 3), (3, 2), (4, 6)\}$

Node coverage $\leftarrow$ No misses N_3

$P_2 = [1, 2, 3, 2, 4, 6, 1, 7] \leftarrow$ miss Node 5

$P_3 = [1, 2, 3, 2, 4, 5, 6, 1, 7]$ No coverage.

Nodes hit: 1-7

edges used:

$\{(1, 2), (2, 3), (3, 2), (2, 4), (4, 5), (5, 6), (6, 1), (1, 7)\}$

Missing edge $\{(4, 6)\} \times$ so ~~have~~

Final answer: test path that has Node coverage but not edge coverage

$P_3 = [1, 2, 3, 2, 4, 5, 6, 1, 7]$ missed edge $\{(4, 6)\}$

- (g) List test paths from the given set that achieve Edge Coverage but not Prime Path Coverage on the graph.

5.6 previously edge coverage = cover 9 edges
 $(1,2), (1,7), (2,3), (2,4), (3,2), (4,5), (4,6), (5,6), (6,1)$

Prime Path Coverage = cover all prime paths

$[2,3,2], [2,3,2]$

Not a simple path: $[1,2,3,2], [3,2,4,6], [6,1,2,3]$

Do test paths

$P_1 = [1,2,4,5,6,1,7] \leftarrow (1,2), (2,4), (4,5), (5,6), (6,1), (1,7)$

$P_2 = [1,2,3,2,4,6,1,7] \leftarrow (1,2), (2,3), (3,2), (2,4), (4,6), (6,1), (1,7)$

Not covered: $(2,3), (3,2), (4,6)$

Not coverage: $(5,6), (4,5)$

$P_3 = [1,2,3,2,4,5,6,1,7] \leftarrow 9 \text{ edges}$

$[1,2,3,2] \quad [3,2,3]$

$[2,3,2,4] \quad [5,6,1,2]$

$[3,2,4,5] \quad [6,1,2,3]$

$[5,6,1,2] \quad \downarrow$

$[2,3,2] \quad \downarrow$

✓

so $P_3 = [1,2,3,2,4,5,6,1,7]$

has edge coverage

But NO prime path coverage

7. Answer questions a-d for the graph defined by the following sets:

- $N = \{1, 2, 3\}$
- $N_0 = \{1\}$
- $N_f = \{3\}$
- $E = \{(1,2), (1,3), (2,1), (2,3), (3,1)\}$

Also consider the following (candidate) paths:

- $p_1 = [1, 2, 3, 1]$
- $p_2 = [1, 3, 1, 2, 3]$
- $p_3 = [1, 2, 3, 1, 2, 1, 3]$
- $p_4 = [2, 3, 1, 3]$
- $p_5 = [1, 2, 3, 2, 3]$

(a) Which of the listed paths are test paths? For any path that is not a test path, explain why not.

7A Test Path

Consist only of valid edges of graph
 $V = \{1, 2, 3\}$
 $E = \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1)\}$
 $N_0 = \{1, 3\}$
 $X P_1 = [1, 2, 3, 1]$ $N_0 = 1 \checkmark$
 ends $Q = 1 \rightarrow X$ Needs to be 3.
 valid edges N_F
 $(1, 2), (2, 3), (3, 1) \leftarrow$ Not test path - ends wrong node.
 $P_2 = [1, 3, 1, 2, 3]$ $N_0 = 1$ edges: $(1, 3), (3, 1), (1, 2), (2, 3)$
 $N_F = 3$ test path $\checkmark$
 $P_3 = [1, 2, 3, 1, 2, 1, 3]$ $N_0 = 1$
 $N_F = 3$ edges: $(1, 2), (2, 3), (3, 1), (1, 2), (2, 1)$
 $X P_4 = [2, 3, 1, 3]$ $N_0 = 2 \times$
 Not test path $(1, 3) \rightarrow \checkmark$ test path
 $X P_5 = [1, 2, 3, 2, 3]$ $N_0 = 1 \checkmark$ edges: $(1, 2), (2, 3), (3, 2)$ Not valid
 $N_F = 3 \checkmark$ Not in
 Not test path
 so... Test paths P_2 & P_3 everything else Not $\checkmark$

7B only length 2 subpaths

(b) List the eight test requirements for Edge-Pair Coverage (only the length two subpaths).

7B

only length 2 subpaths
 $E = \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1)\}$
 Need valid consecutive edge pairs
 For edge $(A \rightarrow B)$ there consists $(B \rightarrow C) = (A \rightarrow B \rightarrow C)$ is valid edge pair
 each
 1. $(1, 2) - (2, 1)$
 2. $(1, 2) - (2, 3)$
 3. $(1, 3) - (3, 1)$
 4. $(2, 1) - (1, 2)$
 5. $(2, 1) - (3, 1)$
 6. $(2, 3) - (3, 1)$
 7. $(3, 1) - (1, 2)$
 8. $(3, 1) - (1, 3)$

7C

Extract edge pairs from test paths

(c) Does the set of test paths from part (a) above satisfy Edge-Pair Coverage? If not, state what is missing.

6c

8: (3,1)-(1,3)

Extract edge pairs from test paths

$P_2 = [1, 3, 1, 2, 3]$ edges (1,3), (3,1), (1,2), (2,3)

$P_3 = [1, 2, 3, 1, 2, 1, 3]$

edges: (1,2), (2,3), (3,1), (1,2), (2,1), (1,3)

were missing
(3,1)-(1,3)

Nope Test paths don't satisfy full Edge-Pair coverage
missing

- (d) Consider the prime path [3, 1, 3] and path p_3 . Does p_3 tour the prime path directly? With a sidetrip?

7d

Path p_3 Prime path [3, 1, 3] test path p_3 [1, 2, 3, 1, 2, 1, 3]

Index	1	2	3	4	5	6
	1	2	3	4	5	6
	1	2	3	4	5	6
	1	2	3	4	5	6
	1	2	3	4	5	6
	1	2	3	4	5	6

No matches to [3, 1, 3]

Let's do side trips

Index: 2: 3
3: 1
4: 5 ← we have a side trip (2 → 1)
6: 3

So 3 → 1 → (2 → 1) → 3

There's a tour w/ a side trip [3, 1, 2, 1, 3]

Path p_3 does tour the prime path [3, 1, 3], but has a sidetrip. Sidetrip [1, 2, 1] between 2 → 3

H.W 4 Pt 5